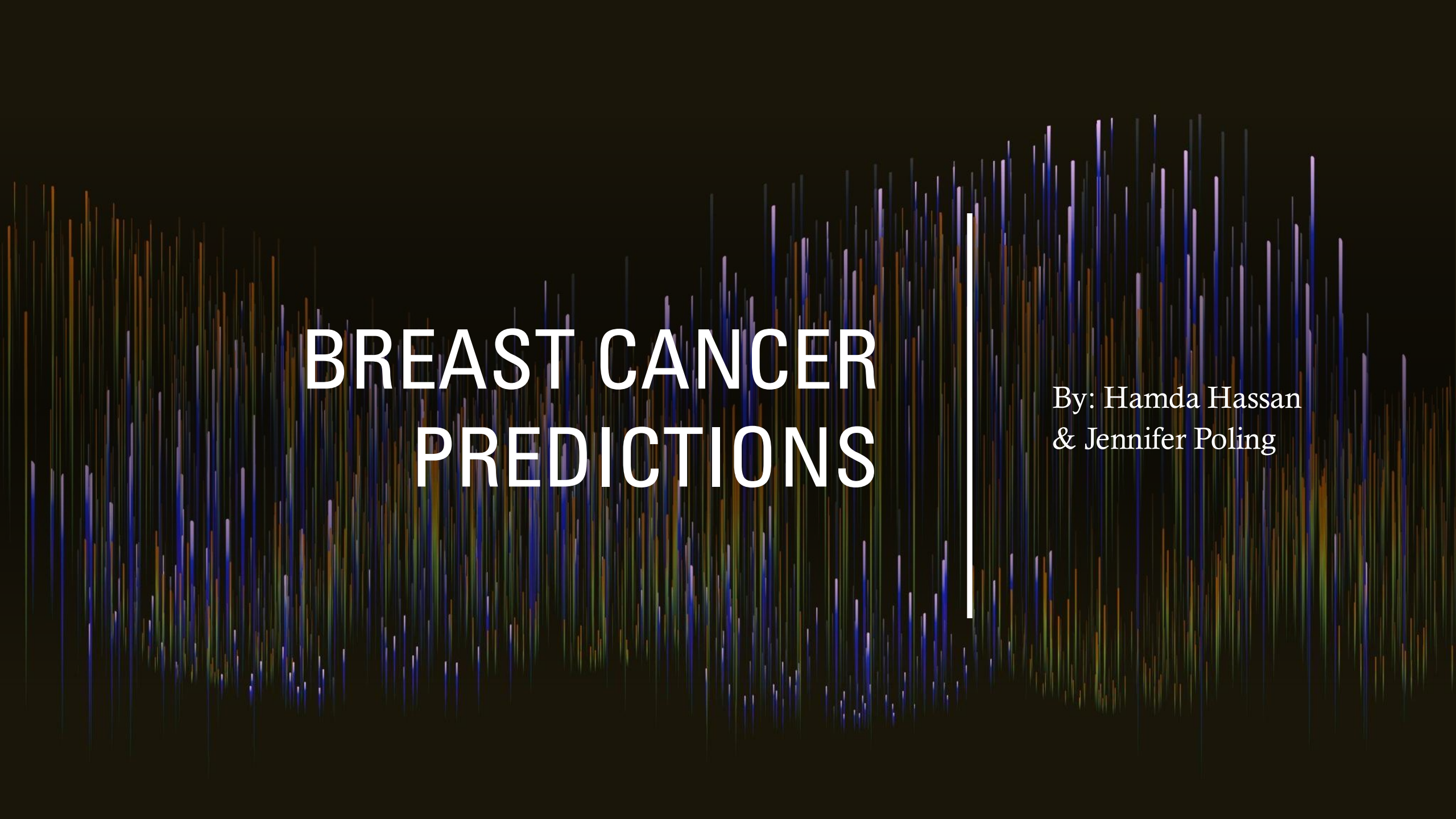




BREAST CANCER PREDICTIONS

By: Hamda Hassan
& Jennifer Poling

AGENDA



Introduction



Data Analysis Process



Findings



Conclusion

INTRODUCTION

1. Dataset Overview

- We worked with the Breast Cancer Wisconsin (Diagnostic) dataset, created by William H. Wolberg together with collaborators Nick Street and Olvi Mangasarian at the University of Wisconsin Hospitals.
- The dataset consists of 30 different variables. These features (radius, concavity, concave points, etc.) describe characteristics of the cell nuclei of the mass.

2. Project Goals

- Can we build a machine learning model to accurately predict whether a tumor is malignant or benign?
 - Which features are the strongest indicators of malignancy?
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DATA DESCRIPTION

Breast Cancer Wisconsin data set

- US Irvine Machine Learning Repository
- Dr. William Wolberg collected samples and performed diagnosis
- Dr. Nick Street and Dr. Olvi Mangasarian digitized and analyzed

DATA DESCRIPTION

Breast Cancer Wisconsin data set

- 30 numeric feature columns
 - First 10 mean of characteristics
 - Second 10 standard deviation
 - Third 10 "worst"
 - 1 diagnosis column (M, B)
 - 569 instances
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DATA DESCRIPTION

Breast Cancer Wisconsin data set

- Informed consent
 - University of Wisconsin Hospitals study
 - Collected for diagnostic purposes
 - Waivers
 - Retrospective, anonymized, compliant with HIPAA
 - No personal information
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DATA PREPARATION

1. **Checked for missing values**

- Verified that all 569 observations had complete data with no null entries.
- Checked for invalid or negative values
- Confirmed that all feature measurements (e.g., radius, area, concavity) were within valid biological ranges.

2. **Considered multicollinearity between features**

- Many features (e.g., radius, perimeter, area) measure similar properties and are highly correlated.
 - We chose robust modeling approaches
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ANALYSIS METHODS

Supervised learning

- Single input logistic regression
 - Multiple input logistic regression
 - Random forest classification
 - XGBoost
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FINDINGS: SINGLE INPUT MODEL (RADIUS 1)

LOGISTIC REGRESSION

Accuracy:

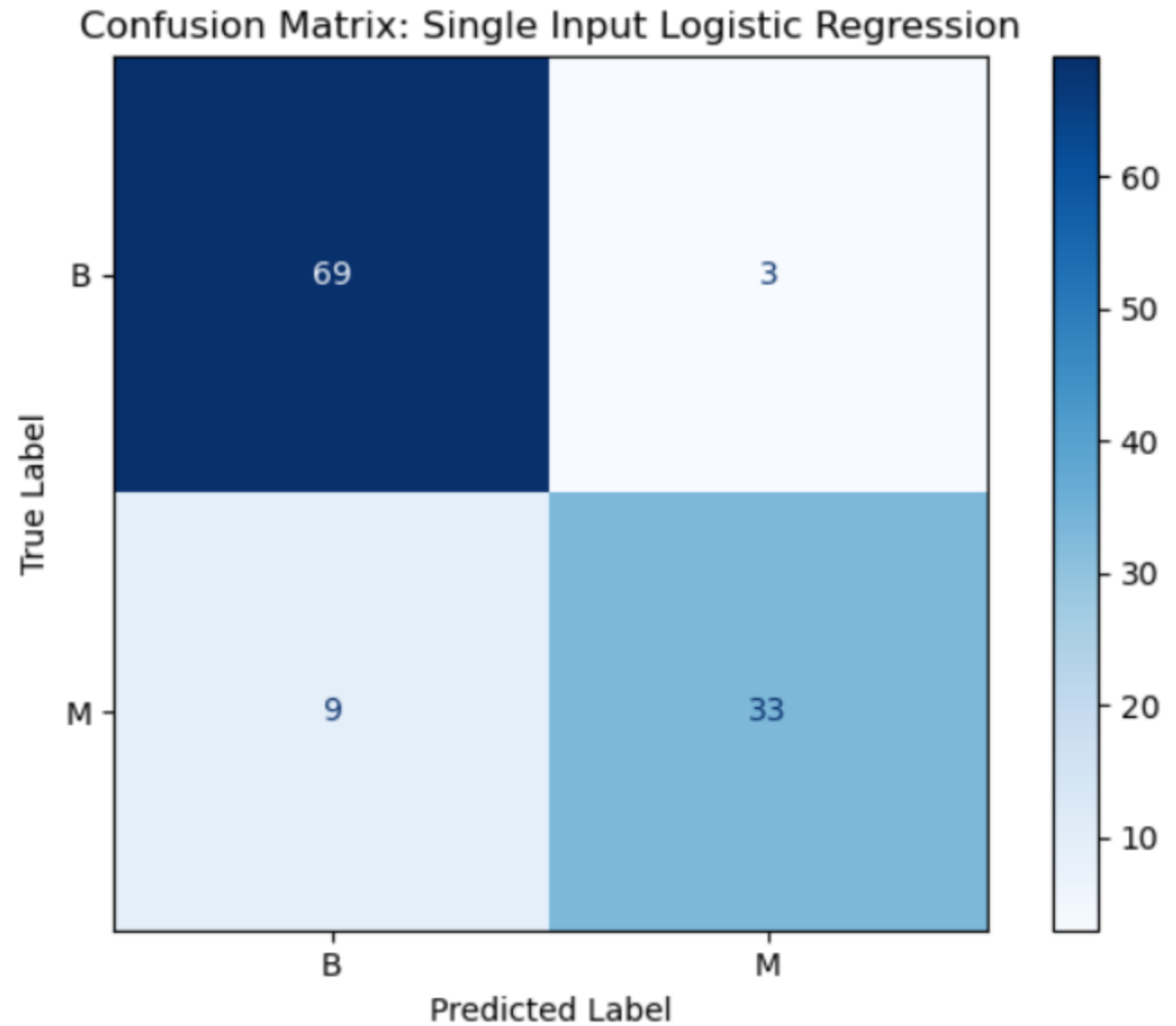
- **89.47%** of the model's predictions matched the actual diagnosis

AUC (Area Under the ROC Curve)

- **94.97%** of the time, the model gives a higher malignancy score to an actual malignant tumor than to a benign one.
-

FINDINGS: SINGLE INPUT MODEL (RADIUS 1)

CONFUSION MATRIX



FINDINGS: SINGLE INPUT MODEL (CONCAVE POINTS 1)

1. Accuracy: 0.6579

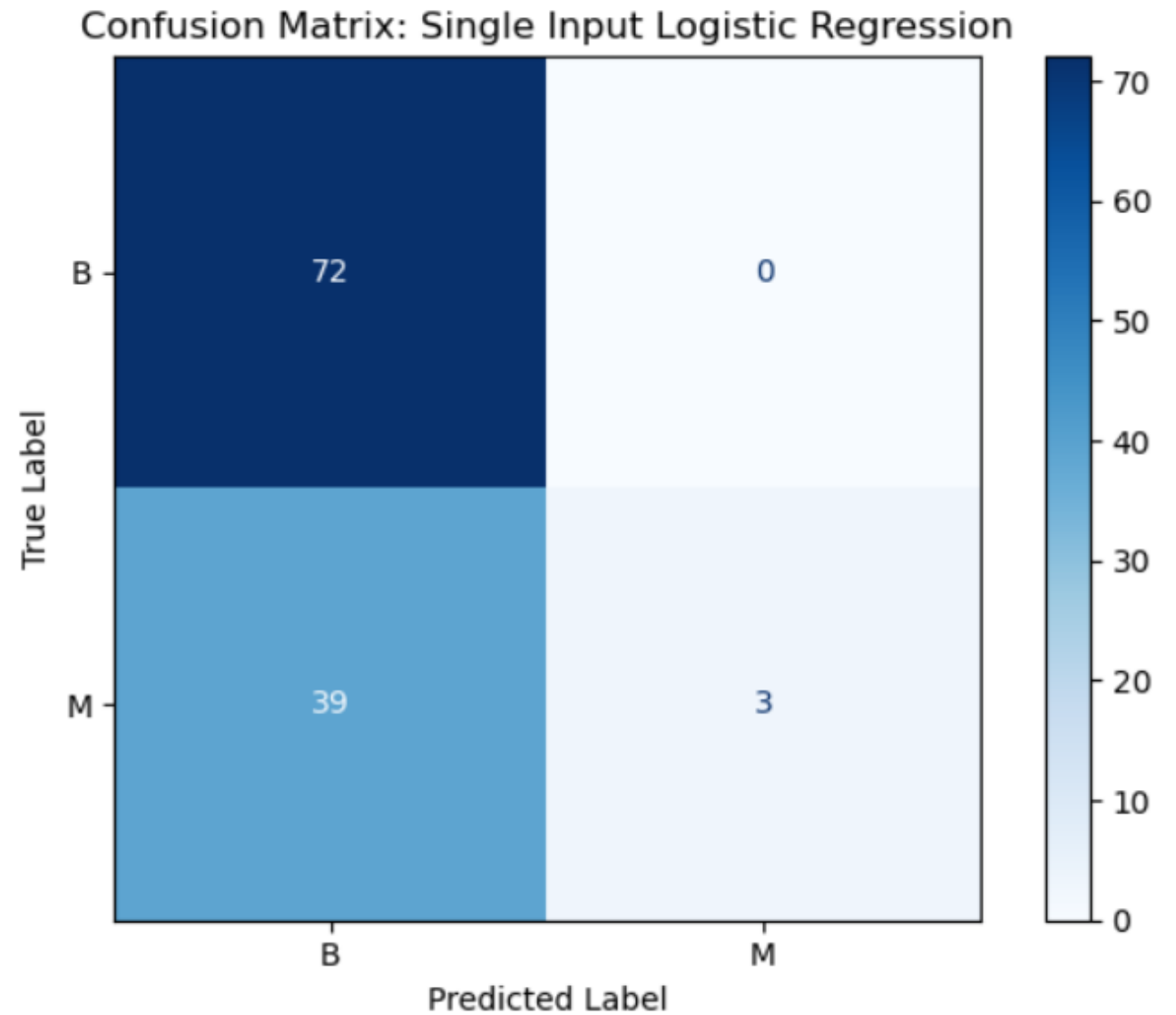
- 66% of the model's predictions matched the actual diagnosis

2. AUC: 0.9676

- 97% of the time the model correctly ranks a randomly chosen patient with the disease as higher risk than a randomly chosen patient without the disease
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FINDINGS: SINGLE MODEL INPUT (CONCAVE POINTS 1)

CONFUSION MATRIX



MULTIPLE INPUT MODEL: LOGISTIC REGRESSION

Accuracy:

- **95.61%** of the model's predictions matched the actual diagnosis

statsmodels p-values: radius, texture, smoothness

multicollinearity: computed Variance Inflation Factors (VIFs)

FINDINGS: MULTIPLE INPUT MODEL LOGISTIC REGRESSION

Accuracy:

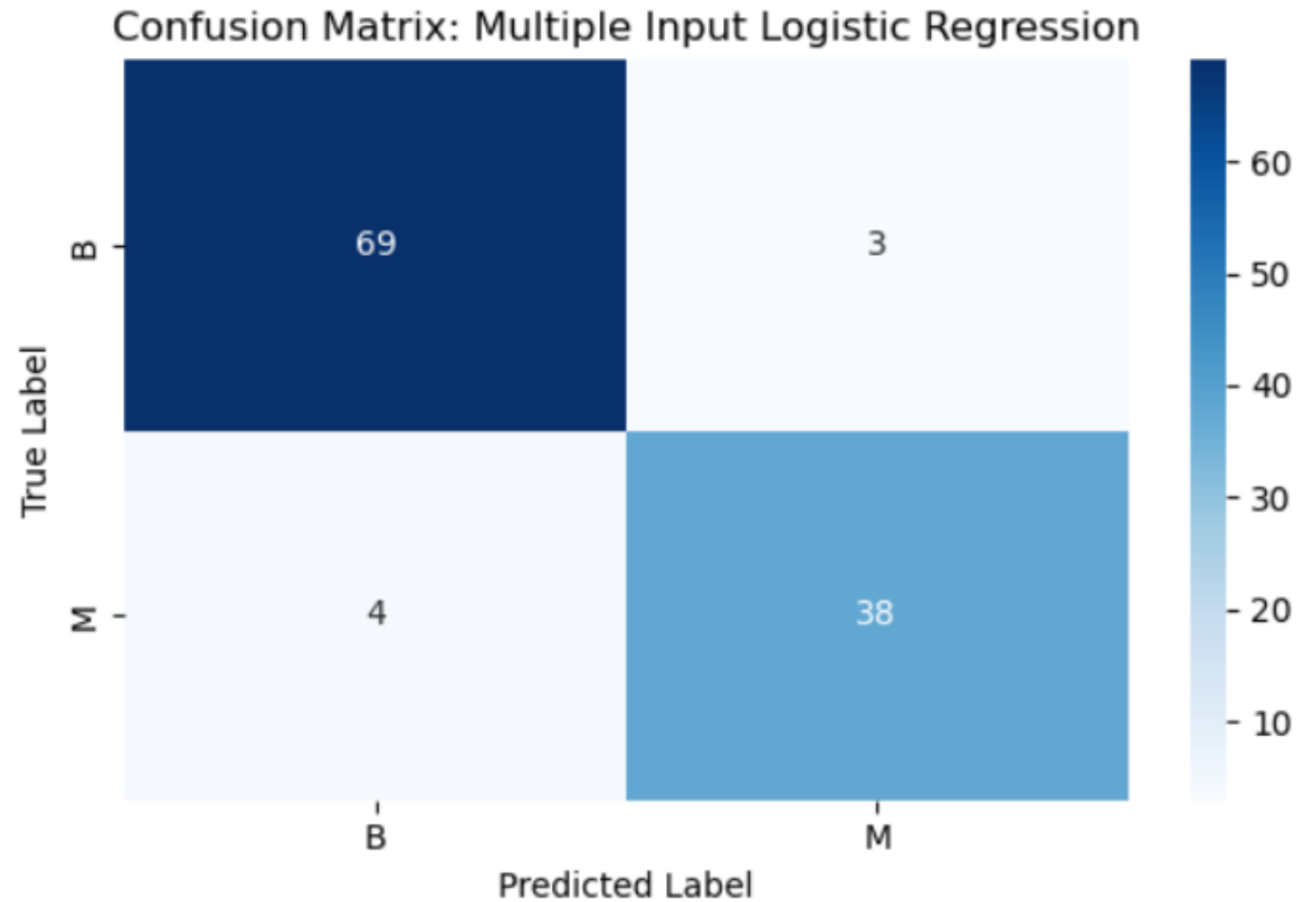
- **93.86%** of the model's predictions matched the actual diagnosis

AUC (Area Under the ROC Curve)

- **93.15%** of the time, the model gives a higher malignancy score to an actual malignant tumor than to a benign one.
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FINDINGS: MULTIPLE INPUT MODEL

CONFUSION MATRIX



FINDINGS: RANDOM FOREST MODEL

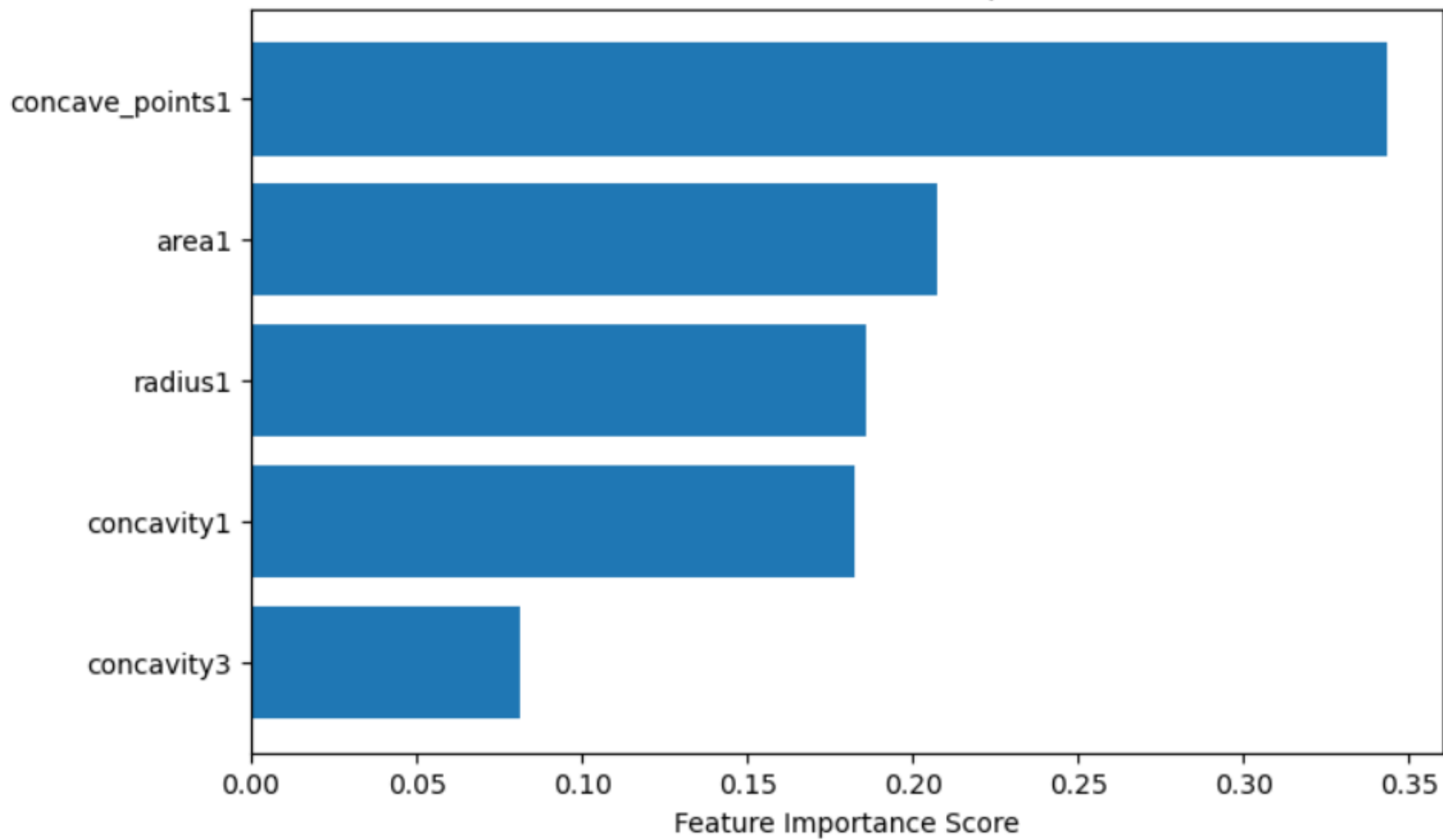
1. Accuracy: 0.9123

- 91% of the model's predictions matched the actual diagnosis

2. AUC: 0.9866

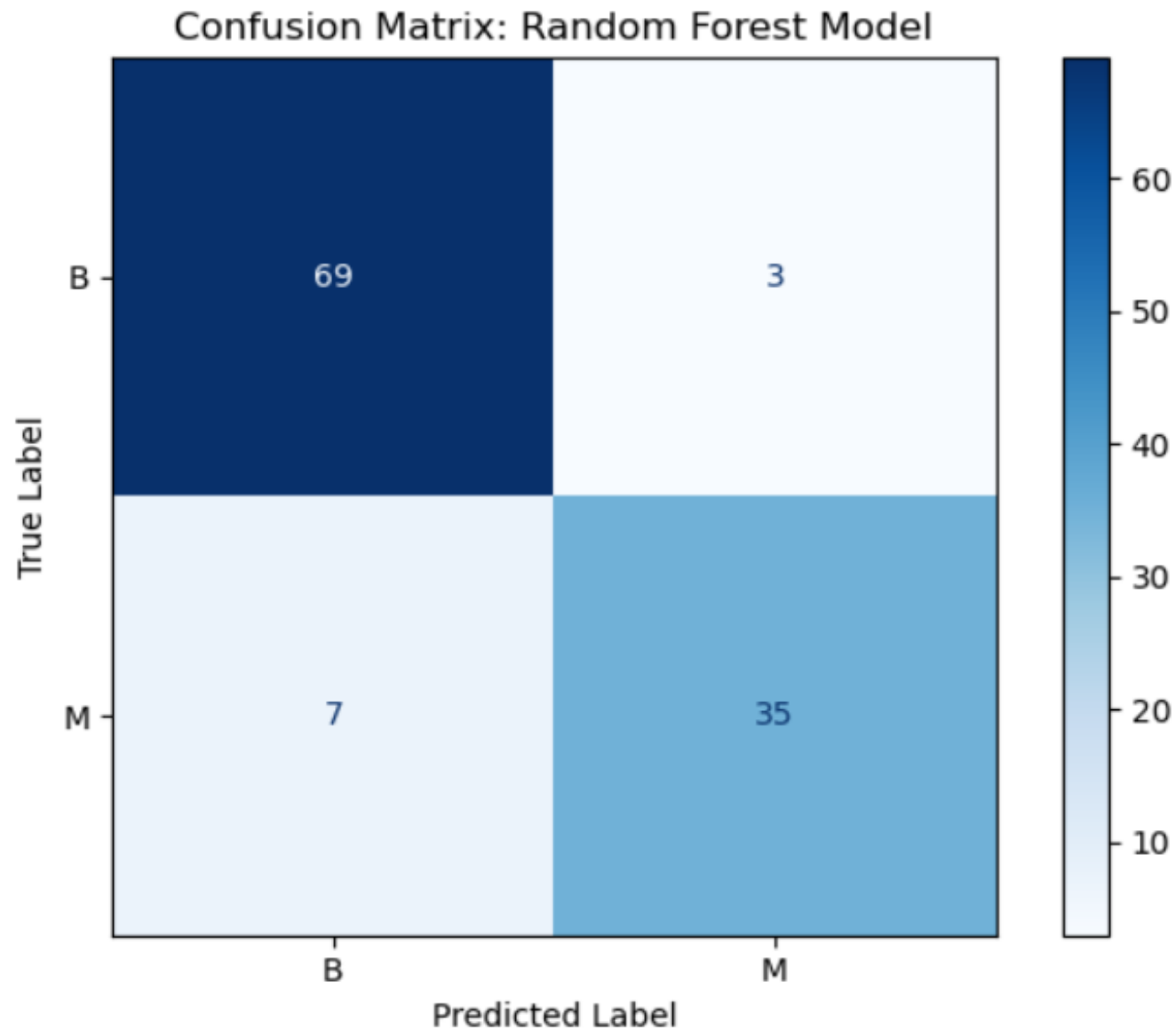
- 99% of the time the model correctly ranks a randomly chosen patient with the disease as higher risk than a randomly chosen patient without the disease

Random Forest Feature Importance



FINDINGS: RANDOM FOREST MODEL

CONFUSION MATRIX



XGBOOST MODEL 1: ALL FEATURES

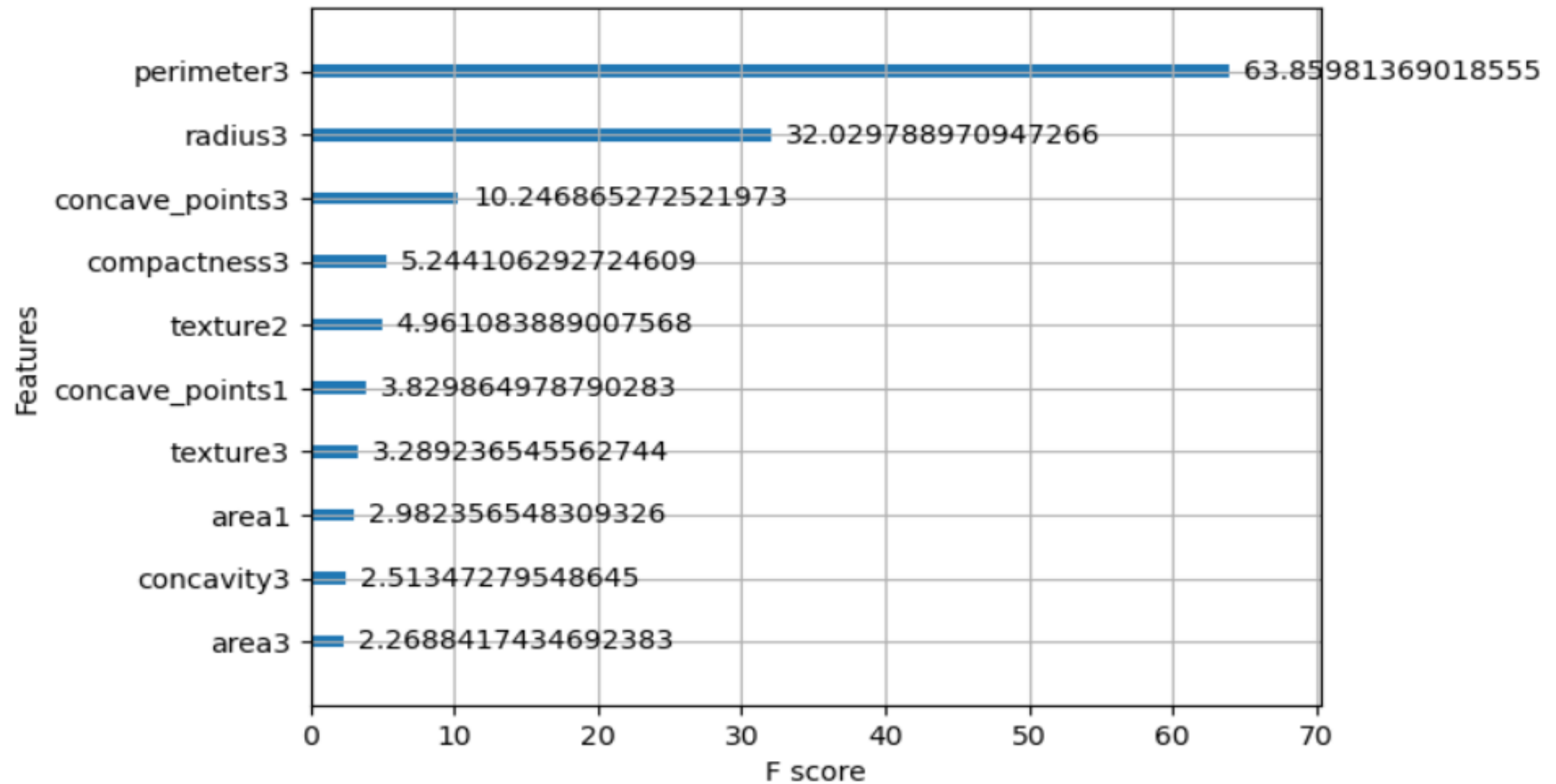
GridSearchCV to find best parameters

Accuracy:

- **98.25%** of the model's predictions matched the actual diagnosis

Same or better accuracy with fewer features?

Top 10 Feature Importances (by gain)



XGBOOST MODEL 1: 10 FEATURES

Recursive Feature Elimination (RFE)

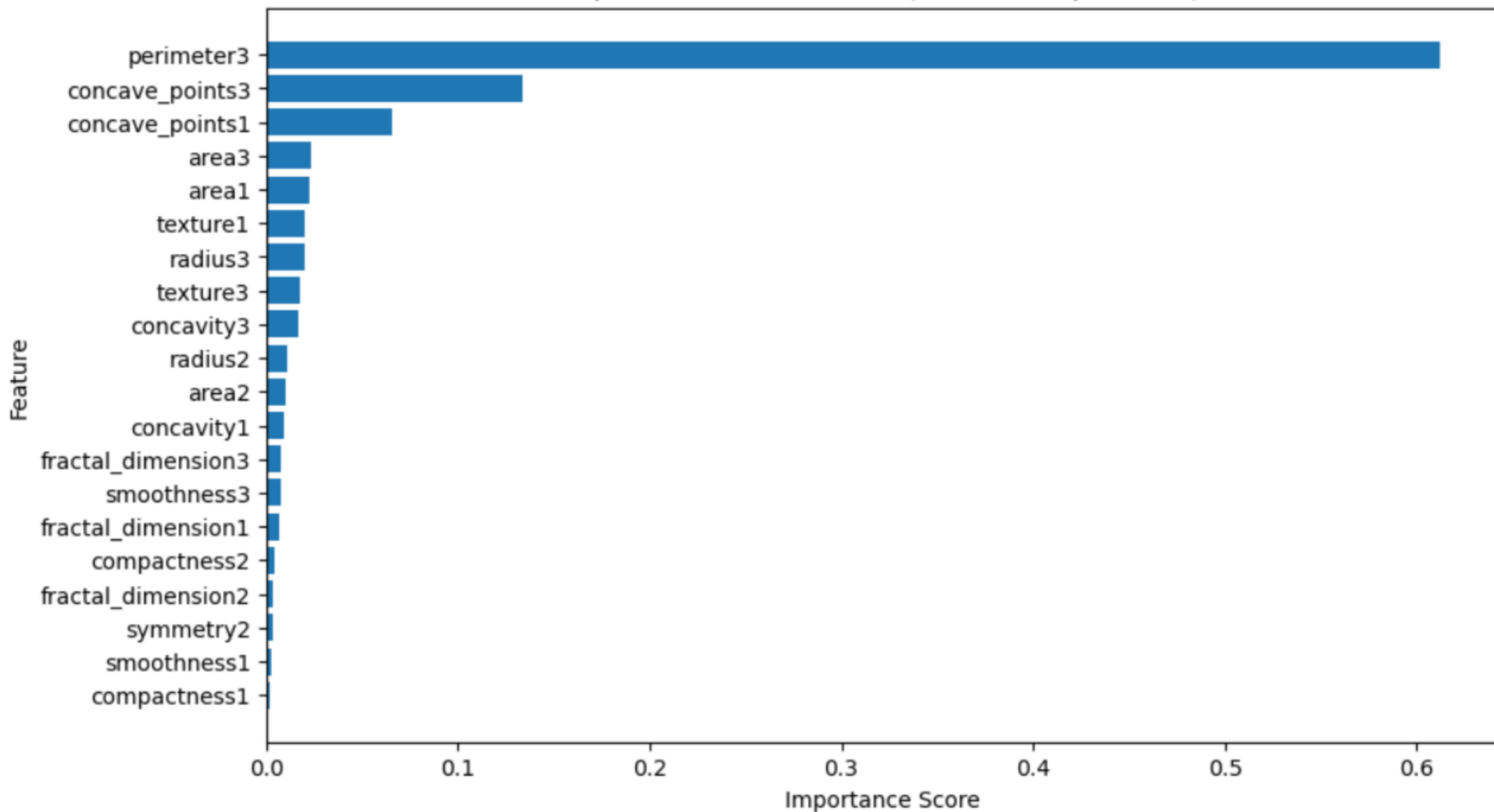
- automatically test which subset of features gives the best accuracy

Accuracy:

- **94.74%** of the model's predictions matched the actual diagnosis

Find optimal number of features

Top 20 Selected Features (XGBoost Importance)



FINDINGS: XGBOOST MODEL

Accuracy:

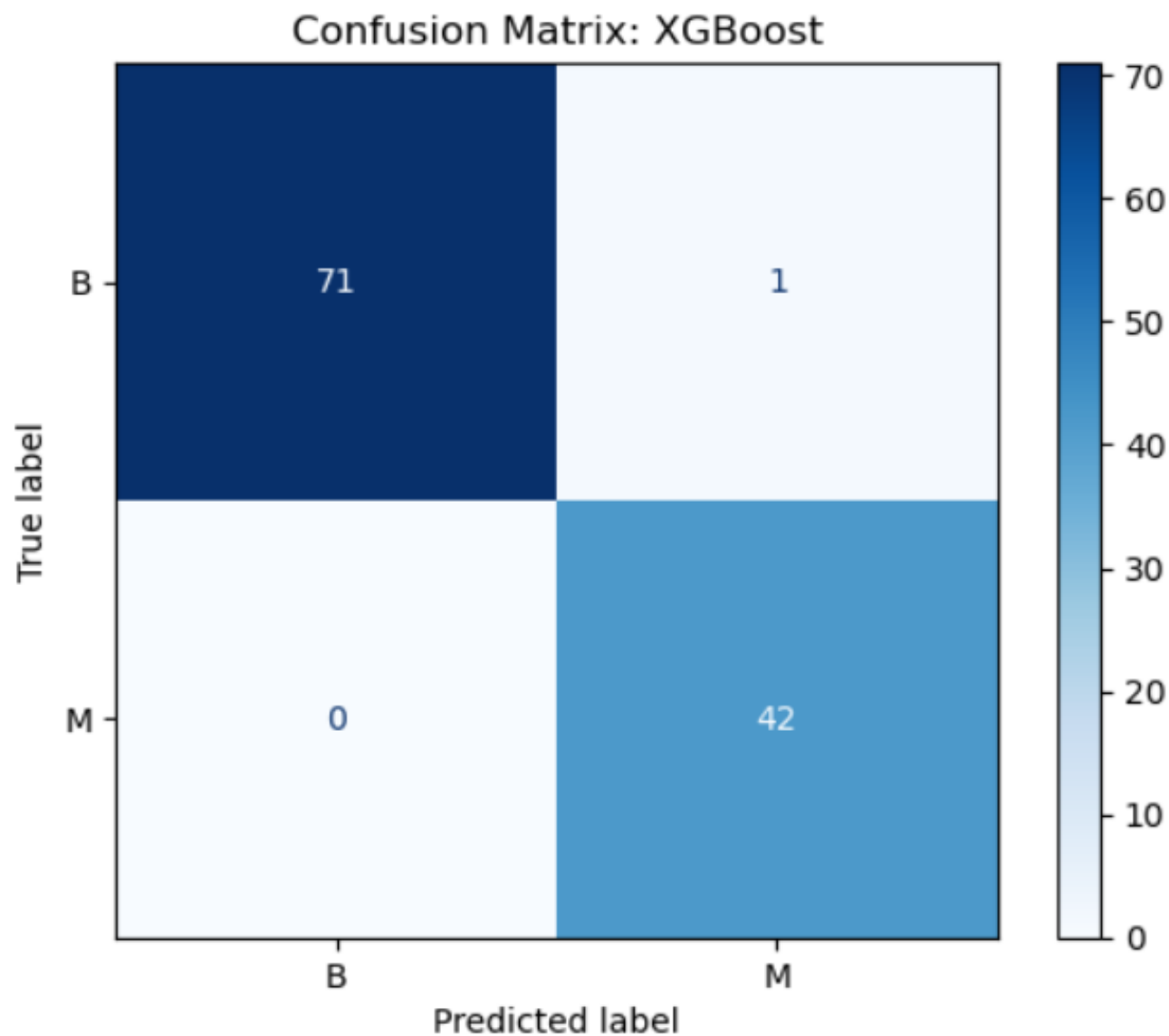
- **99.12%** of the model's predictions matched the actual diagnosis

AUC (Area Under the ROC Curve)

- **99.31%** of the time, the model gives a higher malignancy score to an actual malignant tumor than to a benign one.
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FINDINGS: XGBOOST MODEL

CONFUSION MATRIX



CONCLUSION

- **Features predict cancer quite well**
 - **Best features varied by model**
 - **Best performance: XGBoost model**
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THANK YOU!!!